

Real-Time Eye-Blink and Head Movement Detection for Attention Analysis in Online Learning Environments

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Abstract

Tracking attention for students in an online-based learning environment is very important in order to increase their engagement and learning process. The current research paper proposed a non-invasive attention estimation model using a webcam-based approach based on eye blink dynamics using MediaPipe and Eye Aspect Ratio (EAR). Unlike existing methods that use eye blink frequency, this proposed method uses eye blink duration and defines it as short, normal, and long using fuzzy logic to detect small differences in cognition. In addition to this, head pose and interaction are also considered to increase robustness. Experimentation has shown that eye blinks do not linearly depend on learning performance, and a multimodal approach is required for this purpose. The classification of attention (high, medium, low) was evaluated using deep learning models, and it was found that LSTM resulted in the highest accuracy of 92.33%, followed by GRU (91.48%) and RNN (85.23%) compared with CNN and ANN. It can be deduced from the results that the incorporation of blink dynamics and interaction characteristics with head movement and interaction properties improves the estimation of attention. The proposed system would help in the real-time, scaled, and objective tracking of the activities of the learners on MOOCs and e-learning systems. Future work would include the integration of gaze tracking, emotion recognition, and physiological signals such as EEG for better accuracy.

Keywords: Eye Aspect Ratio, Learning, Massive Open Online Courses, Behavior

1 Introduction

Blinking is a fast neuromuscular response that is required to keep the eyes moist and guard against

the eyes, yet it is also a significant behavioral cue in Human-Computer Interaction (HCI) [1]. In addition to its physiological purpose, blinking is

a non-invasive way of learning about user behavior, especially attention and engagement. It has been demonstrated that the features of blinking are strongly linked to thinking, including attention, working memory, and mental workloads, and hence they are cognitive processes, as well as to mental processes and problems, among others, e.g. memory, learning, etc. [2], [3]. The changes in the blink rate can also indicate cognitive demand; e.g., the frequency of blink usually slows down when the task is very demanding and requires attention to be paid in order not to miss the most important visual data, whereas spontaneous blink can signal the presence of the cognitive process or understanding. In this way, blinking is an attentional state that is sensitive and dynamic. Blink monitoring has been proposed as a potential method of evaluating the engagement of learners in online education settings, especially MOOCs. Recent developments in machine learning and computer vision allow the proper detection of blink with ordinary webcams, which makes them viable and scalable systems of this nature. Blink behavior is very task-specific: in reading, blinks are likely to be made at syntactic pauses, in video-viewing, at the transitions between scenes or the low-information parts of the video material. Also, the blink duration, which is usually between 100 and 400 milliseconds, further describes cognitive and physiological conditions, where longer blink duration can be taken to be fatigue or heavy mental conditions [8], [9]. Blink frequency and duration can be combined to gain a more detailed insight into the user attention during dynamic learning.

These methodologies have their basis on the foundational research in eye tracking and blink analysis and the baseline statistics used to interpret observed behaviors, including average blink rates of 14 to 20 blink per minute, are the baseline statistics as a context to understand the behavior of the participants in the study [11]. Following these innovations, this paper presents a real time system to identify the frequency and duration of a blink using a refined EAR methodology with seven vertical and one horizontal landmark distances, and an enhancement with the use of smoothing and adaptive thresholding. The system involves the fuzzy classification of the types of blink (short, normal, long) and adds head movement analysis so as to differentiate between

focused and distracted states. Connecting these behavioral signs with the performance on the quiz, the proposed solution presents a non-invasive and efficient way to measure the attention of students and facilitate adaptive learning practices. The suggested system identifies the number of eye blinks per minute (which is in the range of 20 blinks) through Approach A and B that is based on the fuzzy logic, classifying the blink scenarios into a set of predefined classes. as is detailed in the next section. It uses eye-blink in conjunction with head pose (pitch, yaw, roll) to approximate levels of attention in real time in an online MOOC learning environment. The calculated attention is confirmed through the comparison of the participant feedback and quiz performance, which be the ground truth of every human being.

2 Literature Survey

Blinking is a fundamental physiological event essential for ocular health, yet its potential as a non-invasive metric for cognitive monitoring has only recently been fully realized [12]. While early Human-Computer Interaction (HCI) systems established the baseline for using voluntary eye closures as control signals [1, 4, 32], the shift toward remote learning during the COVID-19 pandemic accelerated the development of webcam-based engagement tracking. Initial approaches relied on geometric features derived from dlib facial landmarks to compute the Eye Aspect Ratio (EAR) [12–14], with subsequent improvements such as Modified EAR (m-EAR) and EAR Mapping (EARM) addressing limitations related to static thresholds and individual variability [16]. These methods enabled the detection of strong blinks associated with drowsiness [17]. To further enhance robustness, recent studies have incorporated MediaPipe to extract 3D facial landmarks, enabling accurate blink detection under varying head poses and lighting conditions [21, 25]. In parallel, research has emphasized the importance of temporal blink dynamics, where duration and frequency vary with cognitive load and task demands [5, 26]. Signal processing techniques such as Savitzky-Golay filtering and dynamic thresholding have been employed to reduce noise [2], with typical blink durations defined between **100 ms and 400 ms** to distinguish natural blinks from noise and drowsiness-related events. These refined

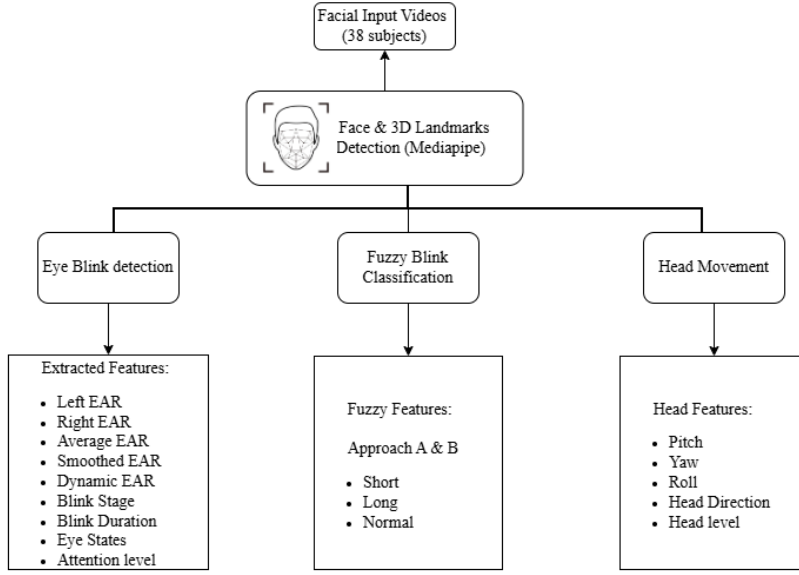


Fig. 1 Eye blink detection system design

metrics have been widely applied in domains such as driver monitoring [9] and industrial settings [5], and are increasingly used to quantify user engagement in digital environments, where synchronized blinking patterns reflect underlying cognitive processes [8, 10, 33].

The integration of these advancements into online education has led to a shift from simple blink detection toward multimodal attention analysis. Recent studies have leveraged blink-based features to detect fatigue and optimize engagement in edge-based learning systems [22, 33]. Datasets such as mEBAL2 and IMPROVE have been utilized to study the effects of distractions, including mobile phone usage, on learner attention [29]. Furthermore, modern frameworks combine multiple visual cues—such as head pose, gaze, and facial expressions—within deep learning models to predict attention levels and group concentration in e-learning environments [30–32]. Advances in Head Pose Estimation (HPE) using deep learning have further improved orientation tracking in unconstrained settings [34]. Current research trends emphasize multimodal systems that integrate head pose with eye tracking and facial behavior to identify complex activities such as distraction or note-taking, thereby enabling a more

comprehensive and reliable assessment of learner engagement [35].

3 Materials and Methods

This section describes the experimental protocol, data collection, and the overall framework adopted for attention analysis.

3.1 Data Collection

The research proposed is aimed at identifying the attention of learners in MOOC settings as a blend of eye-blink dynamics, head pose movements, and quiz performance. It is based on previous research, where variability in blink length and insensitive cognitive variations are managed with the help of a fuzzy-based system, based on earlier research results [29, 30]. The respondents were 38 participants of SSN college (14 male, 24 female, UG, PG, research scholar, and one school student), whose data were recorded with their consent using a Logitech webcam. The participants watched pre-selected lecture videos (theory/problem-solving), and could see some content of puzzles and entertainment as an option, and had to take a mandatory quiz at least after watching one of the videos. Real-time feature detection was done with an

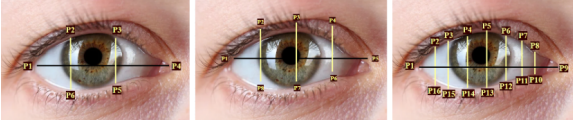


Fig. 2 Eye Aspect Ratio Formula

optimized EAR algorithm to calculate the blink frequency and duration with smoothed EAR and dynamically adjusted thresholds as short, normal, and long blinks. Attention changes and distractions were identified through head pose estimation pitch, yaw, roll. These aspects were combined to produce levels of attention and plotted against quiz scores to produce labeled data to train and test machine and deep learning models used to classify attention.

3.2 Eye Blink Architecture

The proposed system architecture examines the attention of the learners through an integrated eye-blink dynamics, head pose estimation, and quiz performance. Experimental tasks are recorded on facial videos which are processed in real time to extract features of eye and head movement, with eye-blink detection being done by means of a refined Eye Aspect Ratio (EAR) to estimate the onset, duration, and frequency of the blink. Blinks are further divided into short, normal and long durations in order to record fine grain attentional differences with smoothed EAR values and dynamic thresholding implemented to increase robustness. At the same time, head movements and directional attention can be detected with the help of head pose estimation based on the angle of pitch, yaw, and roll, which allows identifying distraction and off-screen behavior. The features of eye-blink and head pose extracted are combined to calculate continuous attention scores and categorical attention levels, which are then overlaid to quiz performance scores as class labels. Training and evaluation of machine learning and deep learning models to classify attention and analyze concentration are done on the resulting labeled dataset.

3.3 Eye Blink Detection Using Eye Aspect Ratio and Mediapipe

Eye Aspect Ratio (EAR) is a geometric measure of eye openness that is computed by calculating the ratio of vertical to horizontal distances between landmarks on the eyelids, a process that is referred to as space relationships. EAR does not change much when the eyes are open, but it decreases significantly during a blink, as the vertical distances become significantly less, and it is therefore a convenient feature to use in blink detection. The Haar cascades and the dlib facial landmark detector are popular methods of eye-blink detection, but these methods cannot resist changes in pose of head, lighting, and fine-grained movements of the eyes. On the contrary, the MediaPipe Face Mesh is embraced in this research because of its highly developed and strong face landmark estimation, which delivers dense and very precise landmark points even in various orientations thus being more adaptable in real-time and unconstrained conditions than the traditional ones.

MediaPipe calibrates the right and left eye horizontal landmark pairs to 33 -133 and 362 -263 respectively and this paper builds on the traditional EAR method by applying all seven of the vertical landmark pairs per eye rather than two or three pairs. The vertical pairs for the right eye include 246-7, 161-163, 160-144, 159-145, 158-153, 157-154, and 173-155, while the left eye uses 398-382, 384-381, 385-380, 386-374, 387-373, 388-390, and 466-249. The suggested approach includes these seven vertical distances, which provides a more detailed representation of eyelid movement, increases sensitivity to the subtle eyelid closures, and increases the accuracy of the blink detection. According to Fig. 2, the proposed 7-vertical EAR configuration demonstrates a finer and more credible estimation of eye closure dynamics compared to traditional 2-vertical and 3-vertical EAR techniques to analyze real-time attention.

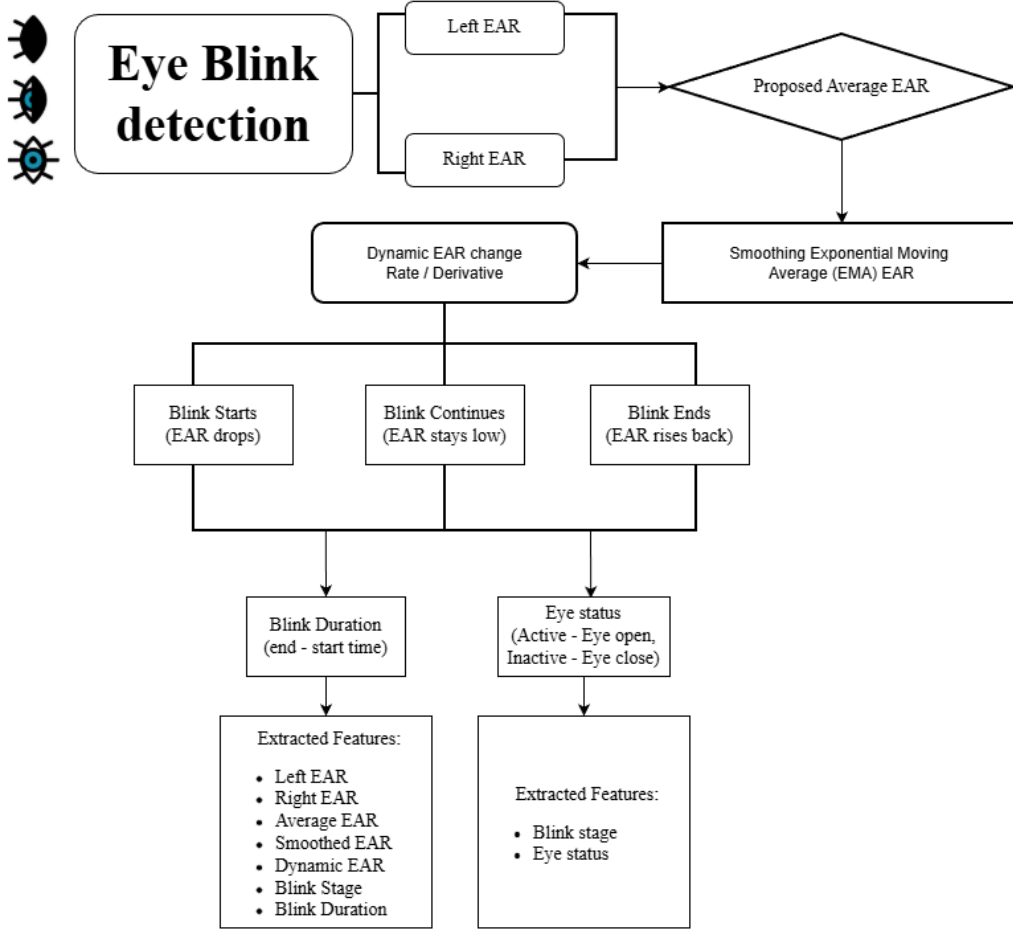


Fig. 3 Feature Extraction from eye blink and head movement

3.3.1 Basic EAR Formula (2 vertical pairs, 1 horizontal pair)

The simplest EAR calculation uses two vertical landmark pairs and one horizontal pair:

$$EAR = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2 \cdot \|p_1 - p_4\|} \quad (1)$$

- (p1) and (p4) represent the horizontal eye corner landmarks.
- (p2) and (p6), (p3) and (p5) are the vertical pairs between the upper and lower eyelids.
- The numerator sums the vertical distances, and the denominator normalizes by twice the horizontal distance.
- This ratio remains relatively constant when the eye is open and decreases sharply during a blink.

3.3.2 Extended EAR Formula (3 vertical pairs, 1 horizontal pair)

To improve robustness, an extended EAR formula uses three vertical pairs and one horizontal pair:

$$EAR = \frac{\|p_2 - p_8\| + \|p_3 - p_7\| + \|p_4 - p_6\|}{3 \cdot \|p_1 - p_5\|} \quad (2)$$

- Here, (p1) and (p5) define the horizontal eye corners.
- Vertical pairs (p2 – p8), (p3 – p7), and (p4 – p6) provide a more detailed vertical measurement.
- Averaging over three vertical distances makes the EAR calculation less sensitive to noise or minor landmark detection errors

3.3.3 Proposed EAR Formula (7 vertical pairs, 1 horizontal pair)

In our research, we extend the EAR calculation further by incorporating seven vertical landmark pairs to capture a comprehensive eye closure profile:

$$EAR = \frac{\|p_2-p_{16}\|+\|p_3-p_{15}\|+\|p_4-p_{14}\|+\|p_5-p_{13}\|+\|p_6-p_{12}\|+\|p_7-p_{11}\|+\|p_8-p_{10}\|}{7 \cdot \|p_1-p_9\|} \quad (3)$$

item Horizontal distance is obtained between (p1) and (p9) which are eye corners. item The numerator adds the distances between seven pairs of vertical landmarks across the eyelid. item This formula takes into account all the seven vertical pairs, thus capturing subtle changes and giving a more accurate and solid measure of the eye openness. item This all-around method enhances the accuracy of blink detection, particularly in dissimilar facial positions and half-blockages.

Equation (1) represents the basic EAR formula, equation (2) shows the extended EAR formula, and equation (3) presents the proposed smoothed EAR formula. These formulas progressively improve blink detection by incorporating additional metrics and smoothing techniques.

3.4 Proposed Method for Smoothed Eye Aspect Ratio (EAR) Calculation and Blink Detection

To ensure stable and reliable blink detection, this study employs an Exponential Moving Average (EMA)-based smoothing technique for the Eye Aspect Ratio (EAR). The smoothed EAR is computed using the formulation presented in Equation (4). This approach helps reduce noise caused by subtle head movements, illumination variations, and landmark detection errors.

$$EAR_{\text{smoothed}} = \frac{EAR_{\text{current}} + (period - 1) \cdot EAR_{\text{previous}}}{period} \quad (4)$$

where:

- EAR_{current} is the raw EAR value obtained from the current video frame,
- EAR_{previous} is the previously smoothed EAR value,

- $period$ is the smoothing factor (typically 10–15).

Relation to the Standard EMA Formula.

The smoothing expression in Equation (4) is derived from the general Exponential Moving Average (EMA) formulation expressed in Equation (5):

$$EMA_t = \alpha x_t + (1 - \alpha) EMA_{t-1}, \quad (5)$$

where x_t is the current observation and α is the smoothing coefficient. To adapt the EMA for EAR smoothing, the coefficient α is chosen as:

$$\alpha = \frac{1}{period}, \quad (6)$$

as shown in Equation (6). Substituting this into the general EMA formulation (Equation 5) yields:

$$EAR_{\text{smoothed}} = \frac{1}{period} EAR_{\text{current}} + \frac{period-1}{period} EAR_{\text{previous}} \quad (7)$$

which is mathematically equivalent to the smoothing model presented in Equation (4).

As seen in equation 7 the smoothed EAR is a combination of a smaller percentage of the current EAR value and a higher percentage of the previously smoothed EAR. As the period gets larger, the value of alpha also decreases and the smoothed EAR will evolve more smoothly. This is a good behavior since it will suppress the short term variation and yet the downwards trends that are meaningful due to real blinks.

Without smoothing, the raw EAR values would be very sensitive to noisy variations, which would cause false blink detections and unstable thresholding dynamics. The formula of the equation (??ema ear) makes sure that only the constant and continual reductions of EAR, usually associated with real blinks, are maintained, whereas random spikes and dips are kept to a minimum. In this way, the smoothing of EMA offers the best trade-off between resistance to noise and responsiveness, enhancing the detection of blink in real-time eye-tracking settings.

The system uses a dynamic EAR threshold which is constantly adjusted according to the smoothed EAR value. Once the EMA based smoothing is done, the threshold is calculated as the product of the smoothed EAR and a ratio

parameter, which ensures that the threshold varies naturally as the eye openness changes gradually. The significance of this adaptive mechanism lies in the fact that fixed thresholds tend to break down when face geometry, illumination, camera distance or user posture varies. The model can be refocused by adjusting the threshold per frame to be more noise resistant and insensitive to individual variations in eye shape. Consequently, blink detection can only be triggered when the EAR has a steady downward movement relative to the user-defined baseline, thereby minimizing false alarms and enhancing the accuracy of real-time blink detection.

3.5 Fuzzy Logic-Based Blink Duration Classification

Blink duration varies significantly across individuals and contexts, making fixed threshold-based classification unreliable [8, 25, 26]. To solve this, a fuzzy logic-based methodology is followed so as to model the inherent ambiguity of the blink behavior by attributing overlapping membership values to short, normal and long blink categories. The procedure allows effective transitions between classes to minimize misclassification due to noise and natural variability and increases robustness with respect to real-time attention analysis. The suggested system uses two configurations; a stricter model (Approach A) when it is necessary to detect the instances exactly and a more flexible model (Approach B) when it is needed to observe the instances in a naturalistic environment. The two methods specify membership functions based on the blink duration, which makes a blink a partial member of two or more categories to assess attention with finer and flexible precision. Table 2 summarizes the fuzzy rules that will be applied in both approaches.

Table 1 Blink duration ranges for fuzzy classification

Category	Approach A	Approach B
Short	$\leq 0.10 - 0.12$	$\leq 0.10 - 0.15$
Normal	$0.10 - 0.40$	$0.10 - 0.40$
Long	$0.35 - > 0.40$	$0.30 - > 0.40$

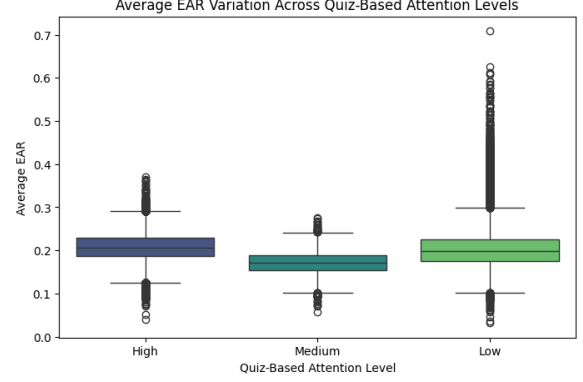


Fig. 4 Blink-Derived Attention Levels Reflected in Average EAR

3.6 Head Pose Estimation for Directional Attention Analysis

Head pose estimation is an important aspect for understanding attentional behavior during online interactions. In the proposed work, head pose is described using three rotation axes: pitch (up/down), yaw (left/right), and roll (tilt), which define head pose directions such as forward, upward, downward, left, and right. MediaPipe’s facial landmark solution is used for head pose estimation because of its high accuracy, computational efficiency, and robust performance under real-time conditions.

Using MediaPipe’s solution for head pose estimation is based on its dense 3D facial landmarks that provide robust head pose estimation even under changing lighting conditions, different facial shapes, and expressions without the need for any special hardware. Using the selected landmarks for a 3D pose estimation model provides robust estimates of head pose in terms of pitch, yaw, and roll axes. A forward pose indicates focused attention, whereas any change in head pose indicates distraction.

4 Experimental Results

However, recent studies show that the normal blink rates vary between 10-15 blinks per minute, with a maximum of 20 blinks per minute under relaxed conditions but reduced while performing tasks [25]. Figure 4 depicts the live blink detection using a Logitech HD 720p camera. In the

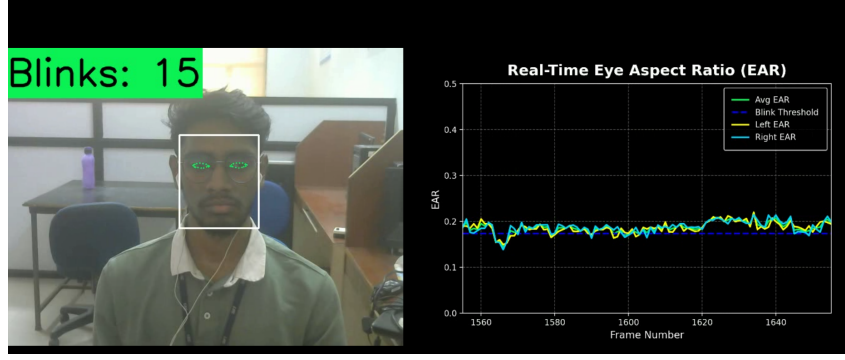


Fig. 5 Real-time eye blink detection, fuzzy classification and head movement

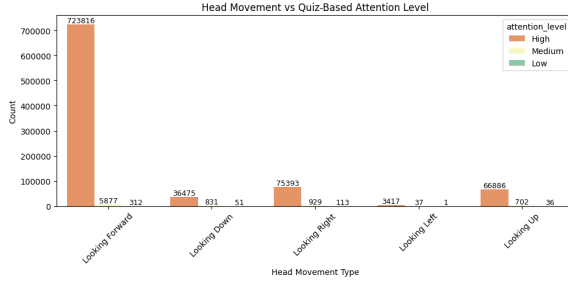


Fig. 6 Head Movement Distribution Across Subjects During Online Learning

case of online learning environments, the exposure time of the screen may cause the blink rates to increase; hence, the blink rates of ≤ 20 per minute are considered normal while detecting the abnormal patterns related to the changes in the attention levels. The total number of participants is 38, resulting in 914,876 samples with low (scores 1-5, 8 participants), medium (scores 6-7, 15 participants), and high (scores 9-10, 12 participants) levels of engagement. The framework proposed in this article incorporates eye blink features such as EAR and fuzzy-based classification with short, normal, and long blink durations along with the head pose features such as pitch, yaw, and roll.

As shown in Fig. 4, the average EAR decreases progressively from high to low attention states, where shorter blink durations (≤ 0.10 s) exhibit higher and more stable EAR values, indicating strong engagement. In contrast, longer blink durations (> 0.40 s) correspond to reduced EAR values, reflecting prolonged eye closure and lower attentional focus. This confirms that EAR is a reliable indicator of cognitive engagement during video-based learning.

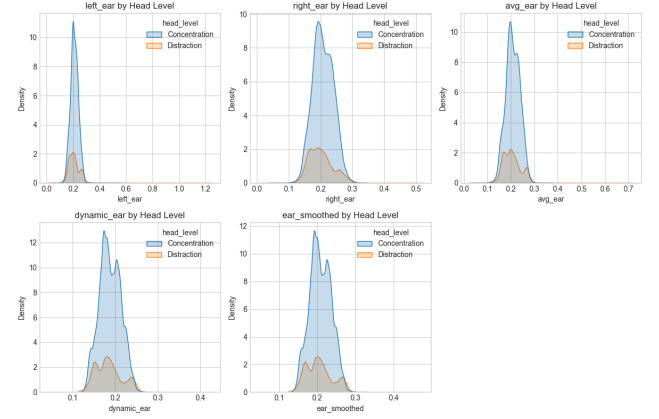


Fig. 7 Density Distribution of Left EAR, Right EAR, Average EAR, Dynamic EAR, and Smoothed EAR Across Concentration and Distraction States

Further, head movement analysis also supports attention estimation using EAR values. In Fig. 6, it is observed that most participants have a predominant Looking Forward (LF) head orientation, indicating their concentration towards the MOOC content. Some lateral head movements, i.e., LL and LR, and vertical head movements, i.e., LU and LD, are minor, indicating participants' distraction during the video. Figure 7 depicts the density distribution of left, right, average, dynamic, and smooth EAR values for concentration and distraction states. In concentration state, peaks are narrower and have higher density, indicating stable eye movements, whereas in distraction state, peaks are broader, indicating unstable eye movements and irregular blinks.

Figure 8 depicts the density distribution of pitch, yaw, and roll head angles for concentration and distraction states. In concentration state,

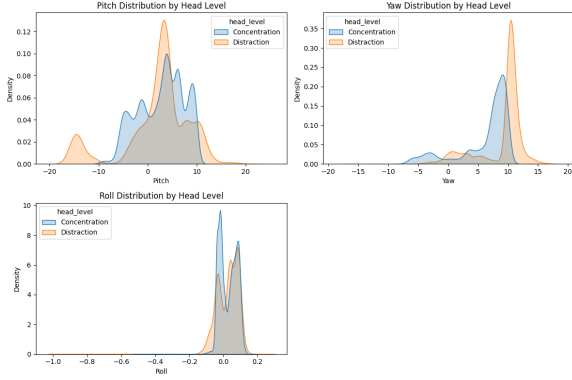


Fig. 8 Pitch, Yaw, and Roll Distribution Across Concentration and Distraction Head Levels

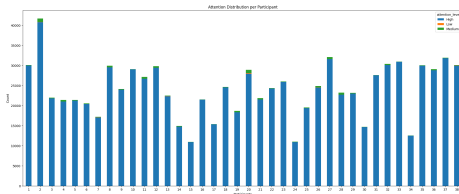


Fig. 9 Blink Attention Level

head angles are closer to zero, indicating stable head movements, whereas in distraction state, head angles are distributed over a larger range, indicating unstable head movements in pitch, yaw, and roll directions. These observations clearly indicate that head movements are strongly correlated with attention state fluctuations. The scatter plot in Fig. 9 depicts the level of attention (High, Medium, Low) among all the participants based on the duration of the blink. The majority of the participants show high levels of attention with a blink duration equal to 0 or ≤ 0.10 s, which is a minimal or partial closure of the eye. The blink duration between 0.10 s and 0.40 s corresponds to medium levels of attention, whereas a blink duration greater than 0.40 s corresponds to low levels of attention due to a prolonged eye closure. Fig. 10 depicts the fuzzy classification results for the participants based on the level of attention with a short, normal, or long blink duration. The majority of the participants show a normal blink duration, whereas a few show a long blink duration corresponding to low levels of attention.

Moreover, the combination of the extracted features using the blink-based approach with the head pose information improves the attention

classification performance. In the case of the high-attention participants, the EAR remains steady, the head is oriented forward, and the blinks are few. In the case of the low-attention participants, the duration of the blinks is increased, the blinks are long, and the head is oriented in different directions. The smoothed EAR and the dynamic thresholding approach minimize the noise in the system. The fuzzy logic-based approach for classifying the blinks is also important since it allows for a smooth transition between the different blink classes. The approach effectively deals with the uncertainty in the duration of the blinks. It is possible to correctly distinguish between different cognitive states. Moreover, the correlation between the attention levels using the blink-based approach and the quiz performance validates the system. It is evident that the system is effective since the participants with the highest attention levels perform well.

Table 2 Performance comparison of deep learning models for attention classification

Model	Train Accuracy	Test Accuracy
ANN	78.13%	81.16%
CNN	82.75%	82.75%
RNN	85.12%	85.23%
GRU	91.57%	91.49%
LSTM	92.26%	92.33%

The performance of these models of deep learning, i.e., Artificial Neural Network (ANN), Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU), was assessed for attention classification at high, medium, and low levels, as depicted in Tables 2 and 3. The input feature set includes left EAR, right EAR, average EAR, dynamic EAR, smoothed EAR, blink duration, and head pose parameters such as pitch, yaw, and roll. The dataset was split into an 80:20 ratio for training and testing with the objective of evaluating the models reliably. Attention is labeled based on a rule-based approach that considers both quiz scores and completion times. Faster responses with higher scores indicate high attention, moderate performance indicates medium attention, and slower responses with lower scores indicate low attention. Deep learning models were used because

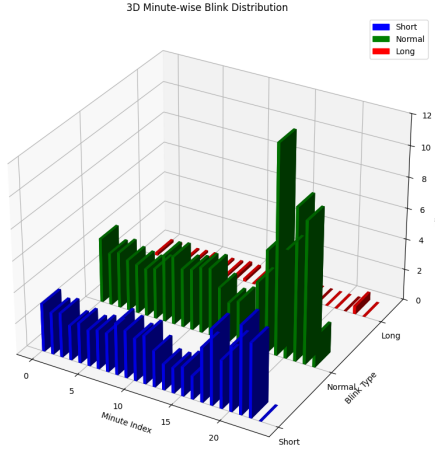


Fig. 10 Fuzzy blink classification

they have the capability to learn complex non-linear relationships and time dependencies within blink dynamics and behavioral patterns. The results showed that temporal models outperform traditional models, as they are able to capture sequential information from the data. Among all models, LSTM showed the highest test accuracy, which is 92.33%, while GRU and RNN showed 91.48% and 85.23%, respectively. CNN and ANN showed poor performance.

Overall, the experimental results confirm the reliability of the suggested approach for the problem of attention monitoring in online learning environments. However, the study is limited by the moderate sample size and experimental conditions. Further research will focus on the investigation of the suggested approach with a larger sample of participants from diverse groups, using different behavioral cues, and developing adaptive real-time systems.

5 Discussion

The results indicate that, although blink duration and blink frequency are very effective parameters for measuring attention, it is not necessarily a linear measure with regard to performance. With the help of EAR-based blink detection using the MediaPipe library, it was possible to accurately measure the blink features and store them in a CSV file; however, the complexity of the behavior of the end user added a layer of complexity with regard to interpretation. For example, it was found that a lower blink frequency equates to a

higher level of attention, whereas a higher blink frequency equates to a lower level of attention, as per the literature. However, it was also found that a high blink frequency equates to a high level of performance with a score of at least 8, whereas a moderate blink pattern equates to a lower level of performance.

In the initial stage, 15 participants were involved in performing tasks such as keyboard usage, resulting in noise and false detection of blinking. To overcome this, a MOOC-based environment was used, focusing on mouse interaction, which is a part of the dataset. Fuzzy logic was used to classify blink duration as short, normal, or long. It was observed that during a glance away from the screen (down, left, right), there were instances of continuous blinking, whereas some users showed less blinking due to mind wandering, resulting in less engagement and lower quiz scores, though less blinking was observed for a period of 2 to 2.5 minutes. Thus, it is observed that blink dynamics, along with head pose and interaction, are reliable for estimating attention. It is observed that blink dynamics alone are not sufficient for reliable attention assessment. A multimodal approach, focusing on blink dynamics, head pose, and interaction, is required for reliable attention estimation.

6 Conclusion and Future Enhancement

The facial video-based detection of blinks can offer a non-intrusive yet effective method for monitoring attention during video-based online learning. By using the duration of the blinks instead of just frequency, the proposed system can effectively detect changes in cognitive states such as attention or distraction. Short and consistent blinks are related to higher levels of attention, while changes in blink duration are related to distraction. Fuzzy logic can also be used to increase the reliability of the system. Additionally, using both blinks and head pose detection can increase reliability over using a single modality. The experimental results have shown that the proposed system can effectively detect attention levels in real time. There are significant correlations between attention levels and learning performance (quiz

Table 3 Class-wise performance (Precision, Recall, F1-score) of deep learning models for attention classification

Model	High			Medium			Low		
	P	R	F1	P	R	F1	P	R	F1
ANN	0.79	0.79	0.79	0.81	0.69	0.75	0.82	0.87	0.85
CNN	0.78	0.84	0.81	0.84	0.73	0.78	0.85	0.86	0.86
RNN	0.82	0.85	0.84	0.83	0.80	0.82	0.88	0.87	0.88
LSTM	0.91	0.91	0.91	0.91	0.90	0.91	0.94	0.94	0.94
GRU	0.90	0.91	0.90	0.91	0.88	0.90	0.93	0.94	0.93

scores). Unlike traditional self-reporting or post-assessment approaches, this method can offer continuous, objective, and real-time monitoring using standard webcams.

Future improvements can also enhance attention detection further using other modalities such as gaze tracking for precise attention allocation to different screen locations, as well as emotion recognition for detecting emotional states such as confusion, boredom, or interest. Moreover, using physiological modalities such as EEG and ECG can provide better insights into the cognitive and emotional states of the user, thereby providing accurate attention detection. Furthermore, using a combination of interaction modalities such as mouse and keyboard interactions, multimodal deep learning models, and adaptive feedback models can provide better support for intelligent tutoring. Moreover, an increase in the dataset with more varied participants in uncontrolled conditions can provide better generalization for the attention monitoring system, paving the way for a comprehensive attention monitoring system using multimodal models for future digital media.

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